

Rave Generation

PRESENTS

TUBEQ



USER MANUAL

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1. Introduction

TubEQ is an analog-style equalizer plugin that combines the musical character of two classic EQ designs in a single, powerful processor. This unique hybrid design features the warm, surgical low-end of the iconic **passive tube EQ** (modeled using Wave Digital Filters), the modern clarity and presence of the **modern air-band EQ** mid/high frequency bands (using State Variable Filters), and the harmonic richness of the **12AX7 tube** from the original passive tube EQ amplifier stage - all finished with a precision brickwall lowpass filter at 24.5kHz.

From the studio-defining low-end warmth heard on countless classic recordings to the crystal-clear "Air" that makes modern productions sparkle, TubEQ delivers the analog sound and character that shaped music history. Whether you're adding weight to kick drums with the famous "boost-and-cut trick," bringing vocals forward with air-band-style presence, or adding harmonic richness with tube character, TubEQ provides the analog sound with the convenience of modern digital control.

Beyond the hardware-style modelling, TubEQ offers the convenience and precision of modern digital control while maintaining the musical, interactive behavior of analog circuits. The result is an equalizer that sounds and feels like the original hardware, with the flexibility and recall of software.

2. Key features

- Passive tube EQ low-end modeling: Wave Digital Filter implementation of a classic passive tube EQ circuit with inductor-capacitor networks
 - Air-band style mid/high bands: Musical frequency points (160Hz, 650Hz, 2.5kHz) plus sweepable Air Band
 - 12AX7 tube character modeling: Modeling of the 12AX7 (ECC83) tube amplifier stage, including subtle harmonic coloration and tonal character
 - The "boost-and-cut trick": Simultaneous boost and cut at low frequencies for enhanced weight and clarity
 - Interactive EQ bands: Analog-style frequency interaction and natural phase relationships
 - Progressive gain compensation: Intelligent level management maintains consistent output while preserving dynamic character
 - Brickwall filter: Precision 24.5kHz lowpass prevents aliasing while preserving musical content
 - Complete analog signal path modeling: From input transformers to output stages, every component modeled
 - Auto makeup gain: Intelligent automatic level compensation maintains consistent output levels while preserving musical character during EQ adjustments
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3. Quick-start

1. Insert TubEQ on your desired track or mix bus
 2. Enable the effect using the "Effect In" switch
 3. Start with the Low frequency section: • Select frequency band (30Hz/60Hz/100Hz)
 4. Shape the midrange and highs: • Use 160Hz for body control • Use 650Hz for vocal presence and punch • Use 2.5kHz for definition and bite • Add Air Band for open, sparkling highs
 5. Add tube character with Tube Input and Tube Level controls for classic tube warmth
 6. Set Output level to match bypassed signal or creative preference
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4. User interface

4.1 Low frequency section (passive tube EQ)

The low frequency section models a classic passive tube EQ using Wave Digital Filter modeling.

Control	Range	Description
LF Boost	0-10	Low frequency boost amount with passive tube EQ curve scaling and progressive response
LF Atten	0-10	Low frequency attenuation with passive tube EQ curve scaling - enables the famous "boost-and-cut trick"
LF Band	30Hz/60Hz/100Hz	Frequency selection switch - both boost and attenuation operate on selected frequency

The Famous "boost-and-cut trick": The original passive tube EQ allows simultaneous boost and attenuation because the circuits operate at slightly different frequencies. This creates a unique

sound where you can add low-end weight while removing muddiness, perfect for kick drums, bass, and full mixes.

Frequency characteristics:

- 30Hz: Deep sub-bass, adds fundamental weight to kick drums and bass
- 60Hz: Classic low-end warmth, excellent for adding body to thin sources
- 100Hz: Upper bass region, great for adding warmth without excessive sub content

4.2 Mid/high frequency section (modern air-band EQ)

The mid and high frequency bands use a modern air-band voicing, with musical frequency points and smooth, continuous gain control.

Control	Range	Description
160 Hz	±5	Low-mid frequency band for warmth and body control
650 Hz	±5	Mid frequency band for vocal presence and clarity
2.5 kHz	±5	Presence band for definition and attack
Air Gain	0-10	Boost amount for the Air Band with progressive curve response
Air Band	off/2.5kHz/5kHz/10kHz/15kHz/20kHz/40kHz	Selects the Air Band frequency - higher frequencies provide smoother, more open sound

Frequency characteristics:

- 160Hz: Controls low-mid warmth and body - cut to reduce muddiness, boost for narrator-style vocal weight
- 650Hz: The "intelligibility" frequency - boost for vocal clarity, cut for a more distant sound
- 2.5kHz: Presence and bite control - crucial for vocal definition and guitar attack
- Air Band: The air-band "Air" - adds open, sparkling highs that enhance perception of detail and space

4.3 Tube section (12AX7 tube stage)

The tube section models a 12AX7 (ECC83) tube amplifier stage, which restores the insertion loss of the passive EQ section while contributing subtle harmonic coloration and tonal character.

Control	Range	Description
Tube In	0-1	Input drive level into the tube stage - higher values increase the tube's tonal contribution
Tube Level	0-1	Amount of tube character processing applied to the signal

Tube modeling characteristics:

While the EQP-1A is not designed as a tube saturator, the 12AX7 does impart subtle harmonic enhancement, especially when driven near the top of its headroom.

TubEQ models these characteristics:

- **Harmonic coloration:** Subtle even and odd harmonics that contribute to the "passive tube EQ sound"
- **Tonal character:** The natural tube response that adds warmth and musicality
- **Headroom behavior:** Response when approaching the +21 dBm maximum output
- **Tube warmth:** The characteristic smoothness and musical quality of the 12AX7

4.4 Output section

Control	Range	Description
Output	±12dB	Final output level control with progressive gain compensation
Auto Gain	off/on	Automatic level compensation based on EQ settings - maintains consistent output levels during EQ adjustments
Effect In	off/on	Master bypass switch - when off, signal passes through clean

4.5 Additional controls

Precise movements: Holding down the 'Shift' key while adjusting a knob will enable you to make precise movements. Allowing for finer control over the audio processing parameters.

Reset knob: Right clicking a knob will reset its value. This action will restore the knob or fader to its default or initial position. Alt + click also works (MacOS: 'option' + click).

Adjust value: Double-clicking a knob allows you to adjust its value directly. This action provides a quick and intuitive way to fine-tune parameters with precision.

Undo/redo: Access Undo and Redo from the menu bar to easily correct or reapply changes. Undo reverses the last adjustment, preventing errors, while Redo restores it, essential for efficient workflow and precise edits.

GUI Opacity: From the menu bar, users can adjust the GUI Opacity from 0% to 100%. This control allows for varying the transparency of the plugin interface, offering better visibility of underlying work areas or personal preference adjustments.

Resize: The interface can be resized from 70% to 200% of its original size via the menu bar. This feature accommodates different screen sizes and resolutions, ensuring TubEQ is accessible and comfortably usable on a wide range of devices and display settings.

Preset menu: Accessible from the menu bar, the Preset Menu provides a comprehensive interface for browsing, loading, and selecting factory presets, enhancing the ease of finding the right settings for any session. Additionally, users can Load/Save, and Copy/Paste presets.

5. Signal flow and processing

Complete signal path

Input → passive tube EQ Low EQ (WDF) → air-band Mid/High EQ (SVF) → 12AX7 Tube Character → Brickwall Lowpass Filter (24.5kHz) → Output Level → Output

Detailed processing chain

passive tube EQ low frequency processing (Wave Digital Filter)

- Inductor-capacitor network modeling
- Passive EQ behavior with tube makeup gain
- Supports simultaneous boost/cut ("boost-and-cut trick")
- Frequency-dependent impedance interactions

Air-band mid/high frequency processing (State Variable Filter)

- 160Hz: Bandpass processing for warmth and body control
- 650Hz: Bandpass processing for vocal presence and clarity
- 2.5kHz: High shelf processing for presence control
- Air Band: Selectable high-frequency enhancement for openness and sparkle

12AX7 tube character (from passive tube EQ)

- Models the amplifier stage tube from the original passive tube EQ hardware
- Subtle harmonic coloration and tonal enhancement
- Natural tube warmth and musical character
- Headroom behavior matching the +21 dBm specification

Brickwall lowpass filter

- 12dB/octave rolloff at 24.5kHz
- Prevents aliasing while preserving musical content
- Maintains phase coherence in the audio band

Auto makeup gain (when enabled)

- Frequency-aware automatic level compensation
- Maintains consistent perceived loudness during EQ adjustments
- Preserves dynamic range and musicality

6. The hardware

Passive tube program EQ

Passive program equalizers of the 1950s used inductor-capacitor networks with a tube makeup gain stage. The famous "boost-and-cut trick" - simultaneously boosting and cutting low frequencies - became a studio standard technique heard on countless classic recordings. TubEQ's Wave Digital Filter implementation captures the complete analog signal path including input/output transformers and the push-pull tube amplifier.

Air-band EQ

Air-band designs pair a few musical fixed frequency points with a sweepable high shelf, lifting detail and openness without the harshness of a conventional treble boost. TubEQ builds this with State Variable Filters and smooth, continuous gain control.

12AX7 tube (ECC83) in the passive tube EQ

The 12AX7 (also listed as ECC83) is part of the amplifier stage in the original passive tube EQ, working alongside the 12AU7 (ECC82) and 6X4 rectifier. While the EQP-1A is not designed as a tube saturator, the 12AX7 does contribute subtle harmonic coloration and tonal character, especially when the unit is driven near its +21 dBm maximum output. This tube's presence is part of what gives the passive tube EQ its distinctive musical character and warmth that users associate with the classic sound.

7. Tips & tricks

The boost-and-cut trick

Start with LF Boost at 3-5 for weight

Add LF Atten at 2-4 to remove muddiness

Result: Enhanced low-end with improved clarity

Perfect for kick drums, bass, and full mixes

Frequency selection guide

30Hz: Deep content - kick drums, bass, full mixes

60Hz: Most versatile - works on most sources

100Hz: Upper bass warmth without excessive sub content

Vocal processing

160Hz: Cut to reduce muddiness, boost for warmth

650Hz: Boost for intelligibility and presence

2.5kHz: Boost for definition, cut to reduce harshness

Air Band: Try 10kHz or 20kHz for open, detailed sound

Air band selection

2.5kHz: Aggressive presence for cutting through mixes

5kHz: Balanced enhancement, versatile for most sources

10kHz: Classic character, smooth and musical

15kHz: Balanced character between 10kHz and 20kHz - musical and smooth

20kHz: Very smooth, adds "air" without harshness

40kHz: Ultra-smooth, subtle enhancement

Tube character

Subtle: Tube In 0.3-0.5, Tube Level 0.2-0.4

Classic: Tube In 0.5-0.7, Tube Level 0.4-0.6

Enhanced: Tube In 0.7-1.0, Tube Level 0.6-1.0

Auto Gain usage

Enable for consistent levels while dialing in EQ settings

Disable for creative level changes or when using TubEQ for parallel processing

Genre applications

Electronic: passive tube EQ for drums/bass, Air Band 10-20kHz, moderate tube

Rock/Pop: boost-and-cut trick on rhythm section, 650Hz for vocals, 20kHz Air

Hip-Hop/R&B: Strong low-end, 650Hz for vocal clarity

Jazz/Acoustic: Gentle enhancement, higher Air Band frequencies (20-40kHz)

8. Installation & troubleshooting

System requirements

- Operating system: macOS 10.13+ or Windows 10+
- Processor: Intel Core i5 or equivalent
- RAM: 4 GB minimum (8 GB recommended)
- Plugin formats: VST3, AU (macOS)

Installation process

1. Run the installer and follow on-screen instructions
3. Locate "TubEQ" in your plugin list under "Rave Generation"

Troubleshooting

- Plugin not appearing: Ensure plugin path is correct in your DAW settings
- Activation issues: Check internet connection and license key accuracy

TubEQ by Rave Generation combines the analog character of a passive tube EQ and an air-band EQ with 12AX7 tube character from the makeup gain stage. Experience the analog sound that shaped music history, now with the convenience and precision of modern digital control.

For more resources, updates, and preset packs, visit ravegeneration.io